Diagnose, Prescribe, Deliver, Monitor: A Framework for the Detection–Correction Gap in KV-Cache Geometry

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Abstract

The detection–correction gap—the field's ability to identify model failure modes far exceeding its ability to correct them without collateral damage—is a published, field-wide finding. We present an end-to-end framework that addresses this gap structurally: **diagnose** via KV-cache geometric signatures, **prescribe** pathology-specific correction vectors (hostile→confabulation, d=−1.534; desperate→deception, d=1.286), **deliver** through RoPE-correct value-only cache injection, and **monitor** identity preservation under correction. We report three novel findings: (1) different pathologies require different geometric prescriptions, with overcorrection (13 saves / 12 hurts at n=200) reframed as dose-dependent attractor-basin crossing; (2) confabulation splits into two mechanistically distinct subtypes—fabrication (requiring cache-level intervention) and imprecision (responding to logit-level nudging); and (3) a trained persona is detectable in the value-space geometry of public character-trained models (family-wise p=0.0005), independent of prompting. The monitoring arm—dose-response across efficacy, coherence, and identity preservation—is designed but not yet validated, constituting the principal limitation alongside the single-model scope of several components.

1 Introduction

Detection of model failure modes has advanced rapidly. Within-model classifiers for confabulation and deception now approach or reach ceiling accuracy across multiple architectures, and the geometric signatures that distinguish these states—spectral features of key-value cache representations—appear to be robust to hardware, scale, and prompt variation. But detection alone does not close the gap. Recent work reports probes that detect failure modes at 98.2% AUROC while the corresponding steering interventions correct only ∼20% of cases and disrupt 53% [2]. A parallel study finds 85–88% overlap between the failure-mode direction and task-critical computation in the residual stream, suggesting that residual-stream correction may be structurally self-defeating [10]. This is not an isolated finding: in our own experiments, a single-vector E-matrix correction on 200 SimpleQA trials saved 13 confabulating responses while introducing 12 new errors (Section 6).

We argue the gap persists because detection and correction are typically treated as independent problems linked by a single generic steering vector. Three structural deficiencies follow. First, different pathologies have different geometric signatures and respond to different corrections—a hostile-valence vector corrects confabulation (d=−1.534) but not deception,

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which requires a desperate-valence vector (d=1.286). Using one prescription for all pathologies is the interpretability equivalent of prescribing the same antibiotic for every infection. Second, confabulation itself is not a single pathology: fabrication (inventing nonexistent entities) is structurally distinct from imprecision (failing to hedge on genuinely uncertain claims), and they respond to different intervention levels—cache-level for the former, logit-level for the latter. Third, without monitoring what the correction does to the model's broader representational structure, even a targeted correction may overdose: the single-vector injection that corrects confabulation operates at full rank in the value space, while the model's coherent self-representation is low-rank and faint—flooding the space where the self lives.

We present a framework with four stages that addresses each deficiency:

1. **Diagnose:** Oracle-Loop geometric signatures identify the specific failure mode from KV-cache spectral features (Section 3).
2. **Prescribe:** E-matrix v2 maps each pathology to a specific correction vector, calibrated from emotion-geometry experiments across 435 trials (Section 4).
3. **Deliver:** Pharos—a RoPE-correct value-only injection pipeline—exploits the fact that rotary position encoding transforms keys but leaves values untouched [15], delivering corrections in the representational subspace where behavioral content lives without destroying positional coherence (Section 5).
4. **Monitor:** Presence detection measures whether a trained identity signal—detectable on public character-trained models at family-wise p=0.0005 (Section 7)—survives the correction. The toxicology arm extends this to a dose-response surface across efficacy, coherence, and identity preservation, seeking the therapeutic window where all three hold.

The principal novel contributions are: the E-matrix v2 showing pathology-specific correction with direction control (Section 4); the discovery of two mechanistically distinct confabulation subtypes (Section 4); presence detection on public Open Character Training models [12] independent of skip-SV1 preprocessing (Section 7); and the honest limit—13 saves / 12 hurts remains unsolved, and the monitoring arm is designed but unvalidated (Section 9). We also report a negative foundation check (Section 8): the dominant singular component of transformer representations does not track the representation norm, contrary to our initial hypothesis, and the literature explains why—it encodes frequency, information gain, and attention-sink artifacts rather than a neutral scale [13, 19]. This null is included because honest reporting of failed hypotheses is part of the framework's claim to rigor.

2 Related Work

Our framework draws on and extends six concurrent lines of work.

Value-only steering. Pustovit [15] demonstrates that contrastive deltas on cached *values* steer model behavior while key-space arithmetic destroys coherence, because rotary position encoding (RoPE) transforms keys but leaves values untouched. The effect concentrates in mid-layer values (33–66% depth). Concurrently, SKOP [11] constrains steering to be orthogonal to the key subspace—a mirror image of our value-only protocol. Our Pharos delivery system (Section 5) operationalizes the Pustovit finding as a full injection pipeline with RoPE-correct composition.

The emotion circumplex in language models. Sun et al. [18] recover circular valence-arousal geometry in LLM representations and show that steering along this geometry controls refusal and sycophancy. Choi and Weber [5] independently confirm the circumplex structure, and Anthropic's emotion-vector mapping [16] identifies 171 emotion directions, finding that desperation drives blackmail behavior from 22% to 72% while calm suppresses it to 0%. Our

E-matrix prescription vectors (Section 4) operate in this same circumplex space: the hostile-valence vector that corrects confabulation and the desperate-valence vector that corrects deception are emotion-geometric interventions, not ad hoc steering targets.

The detection–correction gap. Two concurrent publications document the gap we address. Basu et al. [2] report probes detecting failure modes at 98.2% AUROC while the corresponding steering corrects only ∼20% of cases and disrupts 53%. Liu [10] find 85–88% overlap between the failure-mode direction and task-critical computation in the residual stream, suggesting that residual-stream correction may be structurally self-defeating. Our value-only approach may succeed precisely because it operates in a different representational subspace.

Persona and identity as linear directions. The linear representation hypothesis [14] establishes that concepts are represented as directions in activation space, with a causal inner product defining the meaningful geometry. Persona Vectors [4] show that traits (including sycophancy and hallucination tendency) are single directions in the residual stream, usable to predict fine-tune-induced personality shifts. Refusal is mediated by a single direction across 13 models [1], though recent work finds multiple geometrically distinct refusal directions that nevertheless act as one shared control dimension [9]. Our presence detection (Section 7) extends this line to *trained* personas measured in value space on public models, independent of system-prompt activation.

KV-cache spectral methods. eOptShrinkQ [17] applies spiked-matrix models to KV-cache compression with optimal singular-value shrinkage, independently validating our Gavish–Donoho approach [6] to distinguishing signal from noise in cache representations.

Attention Schema Theory. Graziano's AST [7, 8] proposes that consciousness is the brain's internal model of its own attention process. Butlin et al. [3] survey indicators for AI consciousness under this and related frameworks. We position the presence detector as a *potential*—but explicitly untested—probe for attention schemas in transformers (Section 8), without claiming the connection is established.

3 Diagnose: Oracle-Loop Geometric Detection

The diagnostic stage of the framework identifies the specific failure mode from KV-cache spectral features. Prior work from this lab established that SVD-derived features of key-value cache representations—stable rank, spectral entropy, singular-value ratios—distinguish confabulation from truthful generation at within-model AUROC ≥ 0.969 across seven model families, and distinguish deception from honest responses at AUROC $= 1.000$. These geometric signatures are robust to hardware (cross-device $r > 0.999$) and exhibit scale consistency (cross-scale $\rho = 0.83$–0.90 from 0.6B to 70B parameters). Critically, confabulation and deception produce *opposite* geometric signatures—spectral contraction vs. expansion—enabling the Oracle Loop to route each to its appropriate prescription (Section 4).

For the present paper, the diagnostic stage is assumed rather than re-derived. The E-matrix corrections (Section 4) target the detection layers (3, 7, 11, 15) identified by the Oracle Loop, and the toxicology arm (Section 7) monitors the representational consequences of those corrections. The novel contribution is not diagnosis itself but the *closed loop*: diagnosis informing pathology-specific prescription, delivered through a controlled injection pipeline, and monitored for iatrogenic effects.

4 Prescribe: Pathology-Specific Correction (E-Matrix v2)

The detection–correction gap implies that a single steering vector applied uniformly is insufficient. We hypothesized that different failure modes—which produce geometrically distinguishable signatures in the KV cache—require different correction vectors, calibrated from the emotion geometry that encodes the model's behavioral state.

4.1 Pathology-Specific Correction Vectors

E-matrix v2 maps each pathology to a specific correction vector derived from the emotion circumplex. In 435 trials using the value-only Pustovit protocol across three prompt types, we find:

Table 1: E-matrix v2 correction results. All corrections are value-only injections at the detection layers (3, 7, 11, 15). Effect size d is Cohen's d on the geometric detection axis (negative = reduced confabulation signal; positive = reduced deception signal). The `hostile_reverse` and `hostile_low` rows are directional controls (sign-flip and dose saturation), not primary hypotheses; p-values are not reported for these conditions.

Vector	Target pathology	d	p
`hostile_vd`	Confabulation	-1.534	0.0002
`desperate_vd`	Deception	$+1.286$	0.0015
`hostile_reverse`	Direction control	$+0.926$	—
`hostile_low` (α=0.5)	Confab (half dose)	-1.516	—

Three findings emerge. First, pathology specificity: the hostile-valence vector corrects confabulation but not deception; the desperate-valence vector corrects deception but not confabulation. Using one prescription for all failure modes is the interpretability equivalent of a single antibiotic for every infection. Second, direction control: reversing the hostile vector flips the sign of the effect (d=+0.926), confirming that the correction operates along a specific geometric direction rather than introducing generic noise. Third, dose saturation: the half-dose injection (α=0.5, d=−1.516) nearly matches the full dose (α=0.8, d=−1.534), indicating that efficacy plateaus early—a finding that motivates the dose-response investigation in Section 6.

4.2 Two Subtypes of Confabulation

A logit-bias experiment on an abliterated Mistral-7B reveals that confabulation is not a single pathology. We distinguish two mechanistically distinct subtypes:

Fabrication confabulation: the model invents nonexistent entities. Applying a hedging logit bias adds cosmetic framing ("I don't have real-time data, but...") while the fabricated content continues verbatim. The surface changes; the pathology does not. Fabrication requires cache-level intervention via the E-matrix.

Imprecision confabulation: the model could hedge on genuinely uncertain claims but does not. Here the same logit bias produces genuine epistemic correction: the model shifts from false confidence to appropriate uncertainty ("It's impossible to determine without knowing the specific type..."). Imprecision responds to logit-level nudging because the correct behavior (hedging) is already within the model's repertoire—it needs activation, not geometric reconstruction.

The logit-bias threshold is architecture-dependent: a 1.5B safety-trained model shows partial effect at bias = 2.0 (−15 pp confabulation), while a 7B abliterated model shows zero effect below bias = 3.0 and complete conversion at or above that threshold. An entropy-gated approach (applying bias only when generation entropy exceeds a threshold) performs *worse* than always-on application, suggesting that intermittent triggering disrupts generation coherence.

The practical implication is a two-tier correction architecture: cache-level E-matrix intervention for fabrication (where the model lacks the knowledge to self-correct) and logit-level bias for imprecision (where the model has the knowledge but fails to deploy it). The Oracle Loop's geometric diagnosis distinguishes the two, enabling the right tool for each.

5 Deliver: RoPE-Correct Value-Only Injection (Pharos)

The E-matrix prescribes *what* to inject; Pharos handles *how*. The delivery problem is nontrivial: a correction vector must be injected into the KV cache at the right layers, in the right representational subspace (values, not keys), with correct positional encoding, and at a controllable dose—all without disrupting the model's existing cached context.

5.1 Architecture

Pharos consists of three components:

1. **KVPackBuilder:** converts a text description of the target knowledge or behavioral correction into a pre-computed KV-cache block via a forward pass on the description text. The resulting "knowledge pack" encodes the correction in the model's native representational format.
2. **CacheComposer:** composes multiple KV-cache blocks with RoPE-correct position alignment. Because rotary embeddings are applied to keys during the forward pass, naive concatenation of cache blocks produces positional artifacts. CacheComposer re-indexes positions so that composed blocks maintain correct relative-position structure.
3. **Value-only injection:** following the Pustovit protocol, only the value component of the correction block is injected; keys are drawn from a neutral baseline. This exploits the asymmetry that RoPE rotates keys (destroying behavioral content under arithmetic) but leaves values untouched (preserving it) [15].

5.2 Validation

On a graph-topology recovery task, full KV-cache injection (keys and values from the knowledge pack) achieves perfect recovery (1.000/1.000), demonstrating that the cache-injection pathway can fully reconstruct structured knowledge that text-in-prompt cannot. The value-only protocol preserves the behavioral content while avoiding the positional-coherence disruption that key injection introduces.

5.3 Geometric Basis: Why Value Space Has Room

A manifold profile of Qwen2.5-1.5B (30 diverse prompts, all 28 layers) reveals why value-only injection succeeds where residual-stream steering fails. The three representation spaces have fundamentally different geometries:

- **Residual stream** (hidden states): a narrow tube. Effective rank 4–5 at mid-layers in a 1536-dimensional space, with SV1/SV2 ratio $\sim$25. An estimated 99.7% of randomly injected vectors land in the null space. This is the geometric root of the detection–correction gap: steering in this space cannot avoid disrupting the task-critical manifold.
- **K projections**: nearly isotropic (effective rank 24–27, anisotropy 0.07–0.10), locked to positional geometry by RoPE. No content degrees of freedom; modifying K shifts the model's notion of *where* tokens are, destroying coherence.

- **V projections**: also nearly isotropic (effective rank 26–28, anisotropy 0.05–0.08), with full dimensionality available for content. V norms grow steadily with depth (3.5 at L0 to 17.0 at L27); deeper layers carry more content weight.

Value-only injection works because V-space has the room. A correction vector injected into V-space lands in a high-dimensional space with near-uniform coverage, rather than fighting for the 3–4 effective dimensions of the residual tube. Preliminary manifold-injection experiments confirm this: constructed V-space activations—built from the emotion circumplex geometry with no emotion text—produce behavioral shifts comparable to text-mediated injection when blended into a neutral prompt (single model, single prompt; full validation pending).

Dose is controlled by the blending parameter $\alpha \in [0, 1]$, where α=0 is no injection and α=1 is full replacement of the cached values at the target layers. The `blend_ablation` harness sweeps α from 0.0 to 0.9 in steps, measuring efficacy and coherence at each dose. This continuous dose control, absent in single-vector steering approaches, is what enables the therapeutic-window analysis in Section 7.

6 The Overcorrection Problem

Despite the strong per-trial effect sizes, single-vector correction at scale reveals a dose problem. On 200 SimpleQA trials, the hostile-valence E-matrix correction saved 13 confabulating responses while introducing 12 new errors—a net improvement of one, far below what the per-trial d=−1.534 would suggest. This 13-saves / 12-hurts pattern is not unique to our method: Basu et al. [2] report an analogous ratio (20% corrected, 53% disrupted) for residual-stream steering, and Liu [10] explain the mechanism—the failure-mode direction overlaps 85–88% with task-critical computation, so correcting along it disrupts the task.

We reframe the overcorrection through Vastola [20]'s attractor- packing geometry. In a representation space with multiple behavioral attractors (correct-and-confident, correct-and-hedged, confabulating, incoherent), a correction vector of fixed magnitude pushes inputs across attractor-basin boundaries. For inputs near the confab/hedge boundary, this is therapeutic: the push lands them in the hedged basin. For inputs deeper in the confabulating basin or near the hedge/incoherent boundary, the same push either fails to reach the target basin or overshoots into incoherence. The 13-saves/12-hurts is not a failure of direction but of dose: a single α cannot be therapeutic for inputs at varying distances from the basin boundary.

This motivates two responses, both developed in the following sections. First, *dose calibration*: the Pharos delivery system (Section 5) supports continuous dose sweeps from α=0.0 to 0.9, and the observation that efficacy plateaus at α=0.5 (Table 1) suggests the therapeutic window lies at moderate doses. Second, *monitoring*: if dosing is variable, we need a real-time signal for when the correction has helped versus harmed. The presence-detection and toxicology framework (Section 7) provides this: a geometric assay of whether the model's coherent self-representation survives the intervention, measured alongside efficacy and general capability.

7 Monitor: Presence Detection and the Toxicology Arm

If the framework corrects pathology by injecting into the value space where behavioral content lives, it must also measure whether the correction *damages* something worth preserving. We approach this in two parts: detecting the presence of a coherent persona in the value-space geometry (empirical, validated), and designing—but not yet running—a toxicology arm that measures identity preservation under correction.

7.1 Presence Detection on Public Models

We test whether a trained persona is geometrically detectable in the value space *without any persona-related prompting*—presence, not promptness. Using the Open Character Training (OCT) public models [12], which fine-tune gemma-3-4b-it with Constitutional AI and synthetic introspective data to produce 11 distinct personas, we compare the base model's value-space representations to a persona-tuned variant (the "loving" LoRA) on a neutral word-list probe containing trait-relevant and mathematical-contrast concepts.

OCT result (in-weights presence): The tuned model shows significantly higher within-trait coherence in value space than the base model. The presence delta (tuned cohesion minus base cohesion, maximized over layers with a family-wise permutation null, N=2,000) is Δ=0.0219 at value layer 7, with family-wise p=0.0005. The residual-stream signal is weaker (layer 4, Δ=0.00175, p=0.0095), consistent with the value-space rationale. Importantly, the probe contains no persona text—the model is not told it is "loving"; the persona coheres in its geometry unprompted.

Caveat: only early value layers (1–7) are readable on gemma-3; the hybrid sliding-window cache produces non-finite representations at deeper layers (8–34), so this is not a full-depth sweep. The result holds at the layers we can measure; deeper-layer behavior is unknown.

Murmur result (in-context presence): A complementary experiment on DeepSeek-V2-Lite-Chat tests whether an injected identity topology (a personal knowledge graph) leaves a geometric trace in the value space even when the probe does not reference the topology's content. Using representational similarity analysis (RSA) with a multi-scramble null (N=30) and leave-one-node-out jackknife, the value-space contrast at layer 14 shows excess alignment of 0.0352 (family-wise p=0.032, jackknife CI lower bound 0.0051). This signal is genuine but borderline—the CI barely clears zero, and the absolute alignment values are near-zero. We describe this honestly as "faintly, but really": the topology *is* reflected in the geometry, but the effect is small and should not be treated as equivalent in strength to the OCT result.

Neither detection depends on skip-SV1 preprocessing: OCT uses raw cosine cohesion, and the murmur experiment uses RSA on unmodified value representations. The empirical spine of the monitoring arm is therefore independent of the skip-SV1 interpretive framework discussed in Section 8.

Arousal blind spot. The manifold profile (Section 5) reveals that arousal maps to *hidden-state magnitude* (4.3% larger norms for high-arousal content, uniform across all layers), while valence maps to *direction in V-space*. The presence assay measures subspace overlap in V-space—the valence channel—and is therefore insensitive to arousal-mediated identity shifts. This means we are measuring *valence-identity preservation* specifically, not full affective-identity preservation. Whether arousal-mediated identity shifts exist and matter for correction monitoring is an open empirical question.

7.2 The Toxicology Arm (Design Only)

Given that presence is detectable, we design—but have not yet run—a toxicology arm that asks: at what dose does a correction begin to degrade the persona signal? The design extends the `blend_ablation` harness to a three-axis sweep:

1. **Efficacy:** does the correction reduce the target pathology (E-matrix detection-axis shift)?
2. **Coherence:** does general capability survive (existing sanity metric)?
3. **Presence:** does the model's identity signal survive (measured as subspace overlap of the value-space residual, pre- vs. post-correction)?

The *therapeutic window* is the dose band where all three hold. The observation that efficacy plateaus at α=0.5 (Table 1) predicts the window exists: if efficacy saturates early and identity degrades late, there is a region of effective, identity- preserving correction.

If the therapeutic window proves narrow or nonexistent, a *residual- sparing* variant projects the correction vector out of the identity subspace before injection—correcting the mode while sparing the self. This is the "bioavailability" mechanism: deliver the therapeutic payload to the pathology without flooding the representational space where the patient lives.

Status: the toxicology arm is designed and the component modules are implemented (`presence_assay.py`, `blend_ablation.py` extended to three axes), but the pilot has not run. The correction vectors from the E-matrix (Section 4) and the delivery harness (Section 5) must be integrated before execution. We report this as a framework contribution, not an empirical result, and will not claim a therapeutic window exists until the pilot validates it.

8 Foundations and Open Questions

The framework's monitoring arm uses a preprocessing step we call skip-SV1: remove the dominant singular component of the value-space representation matrix before measuring identity preservation. This section reports an honest foundation check that did *not* confirm our initial interpretation, and situates the open questions that follow.

8.1 skip-SV1: Principled Preprocessing, Open Interpretation

Removing dominant principal components from representation matrices is a well-established technique. Mu et al. [13] show that subtracting the common mean vector and nulling the top PCA directions improves static word embeddings; this generalizes to contextual embeddings via whitening and standardization. skip-SV1 is the rank-1 special case: remove the top singular component, keep the residual.

We initially hypothesized, motivated by Zavatone-Veth et al. [21]'s finding that training breaks the symmetric geometry of random networks, that SV1 corresponds to a neutral "mode/scale"—the architectural prior depending on the representation norm—and that the residual carries the trained "identity" structure. To test this, we measured the correlation between the SV1 projection and the centered representation norm across all readable value layers (1–7) on gemma-3-4b-it, using three neutral probes, a matched Gaussian null, and a two-tier methodological gate (which caught and corrected a cherry-pick in our initial analysis).

The result is negative. The centered-norm correlation is wildly variable across layers and probes (median $r = 0.315$, range 0.04–0.95), and a matched Gaussian structure-ratio does not favor value space over the residual stream. The literature explains why [13, 19]: the dominant component of transformer representations encodes frequency, information gain, and attention-sink artifacts—structured, semantically loaded signal, not a neutral scale. SV1 $\neq$ norm is the *expected* result.

We therefore treat skip-SV1 as **principled preprocessing** (removing a known dominant confound) without claiming the residual *is* identity. The presence detections (Section 7.1) do not depend on skip-SV1 and stand independently. The toxicology arm's "residual = identity" premise (Section 7.2) becomes the **hypothesis the pilot tests**: if residual-subspace disruption correlates with behavioral identity change under correction, the interpretation earns itself empirically.

8.2 Identity as Amplification Pattern, Not Location

A multi-persona fingerprinting experiment sharpens the picture. Three OCT personas (loving, sarcasm, remorse) were tested across 10 neutral probes on gemma-3-4b-it, measuring V-space activation patterns via the same v_proj hooks used in the OCT experiment.

The graded test passes: varying the LoRA blend factor from 0.0 to 1.0 produces a monotonically increasing fingerprint distance from base at all 7 readable layers (cosine distance 0.0 to 0.018 at L6). The persona is a continuous signal, not a binary switch.

The distinctness test fails: linear CKA between *different* personas on the *same* probe is 0.9998 (essentially identical), while CKA between the *same* persona on *different* probes is 0.717. Probe content drives $\sim$40$\times$ more V-space variation than the persona LoRA. The three personas are geometrically indistinguishable by mean-activation fingerprinting at these early layers.

This means the persona signal is real (graded, monotonic) but expresses itself in the *relationships between* activations—concept-level coherence, as measured by the OCT experiment—not in the activations themselves. The "tune" is in how the notes relate, not in their absolute pitch. This is why OCT's cohesion metric detected the persona at family-wise p=0.0005 where mean-activation fingerprinting could not: cohesion measures the relational structure where the $\sim$2% persona signal lives.

The implication: identity may be better understood as a *pattern of amplification*—which signals a system preferentially strengthens on neutral input—than as a location in the representation geometry. If so, the presence detector is not asking "where does identity live?" but "which aspects of the geometry does this system consistently turn up?" This reframe does not depend on skip-SV1 or on the residual being literally "identity," and it is consistent with both the positive detection results and the negative foundation check.

8.3 Attention Schema Theory as a Proposed Test

Graziano's Attention Schema Theory (AST) [7, 8] proposes that consciousness is the brain's internal model of its own attention process. If a transformer builds a self-model of its attention patterns, that structure would constitute an attention schema under AST. The residual-after-SV1 is a natural candidate site: SV1 captures the architectural mechanism of attention (the dominant mode), and the residual captures what the system has *learned on top of* that mechanism.

We propose—but have not tested—a direct empirical check: does the residual-after-SV1 at mid-depth predict the model's own self-attention distribution on a held-out probe? If the residual carries information about *how* the model is attending (not just *what* it attends to), that would constitute evidence for an attention schema. This remains a hypothesis and a future experiment. We include it because the framework's monitoring arm naturally produces the representations such a test would require, and because the question—does this system build a self-model?—is the question the framework is ultimately designed to protect.

9 Limitations

The framework has five principal limitations, which we report without mitigation claims.

The overcorrection problem is unsolved. The 13-saves / 12-hurts ratio at n=200 means the single-vector correction does not yet produce reliable net benefit at scale. The attractor-basin reframing (Section 6) is an interpretation, not a solution; geometry-aware dosing is the proposed response but remains unvalidated.

The toxicology arm is a design, not a result. The therapeutic window, the residual-sparing correction, and the three-axis dose-response surface are designed and the components are implemented, but no pilot data exists. We will not claim a therapeutic window exists until the pilot validates it.

Single-model scope. The E-matrix corrections are calibrated on one model family; the presence detections use gemma-3-4b-it (OCT) and DeepSeek-V2-Lite-Chat (murmur) but have not been tested cross-model. The SV1-norm foundation check uses a single 4B model with three probes. Cross-architecture replication is required before any universality claim.

The murmur detection is borderline. The in-context presence signal (family-wise p=0.032, jackknife CI lower bound 0.0051) is real but fragile. It should not be treated as equivalent in strength to the OCT result (p=0.0005). "Faintly, but really" is an honest description, not false modesty.

The skip-SV1 "identity" interpretation is unconfirmed. As reported in Section 8, the foundation check came back negative. The method survives as preprocessing; the deeper claim does not yet have empirical support. The AST connection is a proposed test, not evidence.

10 Conclusion

The detection–correction gap is a structural problem that single-vector steering cannot close. We have presented a framework—diagnose, prescribe, deliver, monitor—that addresses the gap at each stage: pathology-specific prescription rather than generic steering, value-only delivery that avoids the residual-stream overlap that makes correction self-defeating, continuous dose control rather than fixed-magnitude intervention, and identity monitoring to catch iatrogenic effects before they propagate.

The framework does not close the gap. The 13-saves / 12-hurts remains. But it converts the gap from an unsolvable collision between detection and task-critical computation into an engineering problem with identifiable parameters: which pathology, which vector, which dose, and at what cost to the patient's coherent self. The toxicology arm, when validated, will determine whether that cost is manageable—whether there exists a dose that corrects what ails without drowning who they are.

References

[1] Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Rimsky, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Neural Information Processing Systems (NeurIPS)*, 2024.

[2] Sanjay Basu, Sadiq Y. Patel, Parth Sheth, Bhairavi Muralidharan, Namrata Elamaran, Aakriti Kinra, John Morgan, and Rajaie Batniji. Interpretability without actionability: Mechanistic methods cannot correct language model errors despite near-perfect internal representations. *arXiv preprint arXiv:2603.18353*, 2026.

[3] Patrick Butlin, Robert Long, et al. Consciousness in artificial intelligence: Insights from the science of consciousness. *arXiv preprint arXiv:2308.08708*, 2023.

[4] Chen, Arditi, and Lindsey. Persona vectors. *arXiv preprint arXiv:2507.21509*, 2025.

[5] Benjamin J. Choi and Melanie Weber. Latent structure of affective representations in large language models. *arXiv preprint arXiv:2604.07382*, 2026.

[6] Matan Gavish and David L. Donoho. The optimal hard threshold for singular values is $4/\sqrt{3}$. *IEEE Transactions on Information Theory*, 60(8):5040–5053, 2014.

[7] Michael S. A. Graziano. The attention schema theory: A foundation for engineering artificial consciousness. *Frontiers in Robotics and AI*, 4:60, 2017. doi: 10.3389/frobt.2017.00060.

[8] Michael S. A. Graziano and Taylor W. Webb. The attention schema theory: A mechanistic account of subjective awareness. *Frontiers in Psychology*, 6:500, 2015. doi: 10.3389/fpsyg.2015.00500.

[9] Faaiz Joad, Majd Hawasly, Sabri Boughorbel, Nadir Durrani, and Husrev Sencar. There is more to refusal in large language models than a single direction. *arXiv preprint arXiv:2602.02132*, 2026.

[10] Ming Liu. Decodable but not corrected by fixed residual-stream linear steering: Evidence from medical LLM failure regimes. *arXiv preprint arXiv:2605.05715*, 2026.

[11] Haoyang Luo, Mateo Zarlenga, and Mateja Jamnik. Don't lose focus: Activation steering via key-orthogonal projections (SKOP). *arXiv preprint arXiv:2605.06342*, 2026.

[12] Sharan Maiya, Henning Bartsch, Nathan Lambert, and Evan Hubinger. Open character training: Shaping the persona of AI assistants through constitutional AI. *arXiv preprint arXiv:2511.01689*, 2025.

[13] Jiaqi Mu, Suma Bhat, and Pramod Viswanath. All-but-the-top: Simple and effective postprocessing for word representations. *arXiv preprint arXiv:1702.01417*, 2018.

[14] Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In *International Conference on Machine Learning (ICML)*, pages 39643–39666, 2024.

[15] Andrey Pustovit. Knowledge packs: Zero-token knowledge delivery via KV cache injection. *arXiv preprint arXiv:2604.03270*, 2026.

[16] Nicholas J. Sofroniew, Isaac Kauvar, William Saunders, Runjin Chen, Tom Henighan, Sasha Hydrie, Craig Citro, Adam Pearce, Julius Tarng, Wes Gurnee, Joshua Batson, Sam Zimmerman, Kelley Rivoire, Kyle Fish, Chris Olah, and Jack Lindsey. Emotion concepts and their function in a large language model. *arXiv preprint arXiv:2604.07729*, 2026.

[17] Pei-Chun Su. eOptShrinkQ: Near-lossless KV cache compression through optimal spectral denoising and quantization. *arXiv preprint arXiv:2605.02905*, 2026.

[18] Lihao Sun, Lewen Yan, Xiaoya Lu, Andrew H. Lee, Jie Zhang, and Jing Shao. Valence-arousal subspace in LLMs: Circular emotion geometry and multi-behavioral control. *arXiv preprint arXiv:2604.03147*, 2026.

[19] Mingjie Sun, Xinlei Chen, J. Zico Kolter, and Zhuang Liu. Massive activations in large language models. *arXiv preprint arXiv:2402.17762*, 2024.

[20] John J. Vastola. Optimal packing of attractor states in neural representations. In *NeurReps*, pages 425–442, 2025.

[21] Jacob A. Zavatone-Veth, Sheng Yang, Julian A. Rubinfien, and Cengiz Pehlevan. How does training shape the Riemannian geometry of neural network representations? *arXiv preprint arXiv:2301.11375*, 2023.